

Solution

FRICTION -01

Class 08 - Science

Section A

1. **(b)** Fish and birds
Explanation: Fish and birds have streamlined body to reduce the friction offered by water and air respectively. They have pointed tips and broader middle portion that help in reducing friction.
2. **(a)** solids, liquids and gases
Explanation: Sound needs a medium through which it can travel. Sound cannot travel through a vacuum. Sound can travel through solid, liquid, and gases.
3. **(a)** Force of friction is very less
Explanation: It is difficult to walk on an oily floor because force of friction is very less on oily surface. The irregularities present on surface are filled with oils that prevent the interlocking of the two surfaces.
4. **(a)** Friction between air and supersonic object
Explanation: When a body moves with speed greater than speed of sound cracking sound is produced in air called sonic boom. Sonic boom is due to friction between air and supersonic objects.
5. **(a)** Smoothness of surface and nature of surface
Explanation: Friction depends upon the smoothness of surface and nature of surface. Rough surface offers more friction than smooth surface
6. **(c)** Fill the irregularities in surface
Explanation: Oils and lubricants help in reducing friction because the lubricants fill the irregularities in the surface of moving bodies and hence reduce friction.
7. **(a)** Frictional force
Explanation: Magnetic force, gravitational force and electrostatic force are example of non-contact force while frictional force is an example of contact force because it arise only when two surfaces are in contact to each other.
8. **(a)** Liquids and gases
Explanation: The liquid and gases together called fluids. They have nature to flow due to less intermolecular force of attraction between their molecules.
9. **(b)** Less friction
Explanation: A car with worn out treads on tyres is more likely to skid due to less friction. The treads in tyres produce more friction that prevents the vehicle from skidding by providing more grip on road.
10. **(c)** Air
Explanation: Gases offers less friction than liquid and solid because in air there is more intercellular space. Hence, air will offer less friction in comparison to kerosene, water, and ice.
11. **(a)** Goalkeeper gloves
Explanation: Car tyres, ice skates, and wicketkeeper gloves have more friction as compared to goalkeeper gloves as more friction is not required in this case.
12. **(a)** Friction between ball and grass
Explanation: Moving ball slows down its speed in a grassy field due to friction between ball and grass. Rough surface offers more friction than smooth surface which slows down the speed of a moving ball.
13. **(c)** Treading the tyres of vehicles
Explanation: Using ball bearing, lubricating and greasing reduce the friction between two surfaces while treading the tyres of vehicles increase the friction that is necessary for motion as it increases the contact area which increases the friction.

14. **(d) Moving object**
Explanation: The force of friction always acts on all moving objects due to contact between the two surfaces. Object moving in air, water, or on surface experiences friction.
15. **(c) gravel**
Explanation: Gravel because the contact area is rough thus the friction is more.
16. **(c) treading of tyres**
Explanation: Friction is increased by treading of tyres. Tread does have an influence on traction in road tyres, if it is not so high that it lifts the tread ground off the road (<0.2 mm) helps sink tread rubber into rough road surfaces and penetrate lubricants or dirt. Peaks in the tread increase press into surface services and add contact points between the road and tyre and thus increase friction.
17. **(b) irrespective of whether the objects are solid, liquid, or gaseous.**
Explanation: The force of friction comes into play irrespective of the states of matter. The force of friction does not depend upon the state of matter of any particular object. Force of friction acts in solids, liquids, and gases.
18. **(b) Increase friction**
Explanation: The shoes of football players have spikes in their sole to increase friction with ground while running to prevent slipping.
19. **(c) Kinetic friction**
Explanation: Kinetic friction is the force of friction which comes into play when one body moves over the surface of other body.
20. **(c) Powder is sprinkled**
Explanation: The striker of carom moves longer distance with same force when powder is sprinkled on board. The minute particles of powder fill the irregularities on surface that reduce the friction and striker easily cover more distance.
21. **(b) Friction increases**
Explanation: It is difficult to move a cycle with brakes on because friction between wheels and road surface increases as both the surfaces interlock each other.
22. **(a) Interlocking between surfaces**
Explanation: When we attempt to move any surface, we apply a force to overcome interlocking between the surfaces called static friction present due to irregularities on the surface.
23. **(c) Slows down them**
Explanation: Air resistance slows down the falling object as it increases with the increase in surface area of the object and acts against the gravitational force due to which it slows down the speed of object to fall down.
24. **(a) increase**
Explanation: After prolonged use, the mortar and pestle (silbatta) lose the roughness due to which frictional force reduces and it does not work. So, we have to make it rough again for increase friction.
25. **(a) Converting sliding friction into rolling friction**
Explanation: The moving parts of machine are fitted with ball bearing to convert sliding friction into rolling friction. As rolling friction is always less than sliding friction due to less area in contact..
26. **(a) Using lubricants**
Explanation: In machine, friction between moving part is reduced by using lubricants. These lubricants fill the irregularities of the surface and make it smoother.

27. **(a)** sprinkle talcum powder on the wooden surface.
Explanation: The powder reduces the friction between the surface. So, the ball will go at a longer distance due to less friction.
28. **(c)** Dynamic or kinetic friction
Explanation: The friction develops when one body moves over the other body is called kinetic friction. Rolling friction and sliding friction are two types of kinetic friction.
29. **(c)** Friction
Explanation: The force responsible for changing the state of motion of an object is friction. Moving body slows down due to friction and finally come to rest.
30. **(b)** All of these
Explanation: All of these
31. **(c)** Pointed tips
Explanation: The objects with pointed tips will have minimum friction in a fluid. The pointed tips offer less resistance due to less area of contact.
32. **(a)** Drag
Explanation: The frictional force exerted by fluids is also called as drag. This kinds of frictional force can be overcome by changing the shape of the moving body in fluids. Such shape is called streamlined shape
33. **(b)** Decrease friction
Explanation: Decrease friction
34. **(b)** Friction
Explanation: A force that opposes the motion is called friction. It generates when two surfaces are in contact with each other. Friction depends upon the type of surface whether smooth or rough. Rough surface offers more friction.
35. **(c)** Frictional force is not enough to prevent slip
Explanation: Slippery ground have very less friction to prevent slipping so, it is difficult to walk on such grounds. As it does not allow two surfaces to be in contact with each other properly.
36. **(c)** Heat
Explanation: Friction produces heat. When two surfaces in contact rub each other produces friction resulting in the production of heat. All machines get warm due to this.
37. **(a)** Movement of ships or boat
Explanation: The friction offered by water is good for the movement of ships and boat in water. In ships and boat applied force is directed in backward direction and due to opposite nature of frictional force ships or boat moves in forward direction.
38. **(c)** Making tips thin
Explanation: Streamlining is making the tips thin. Streamlining shape reduce the friction in fluids like air and water as it has narrow tips which reduces the area of contact
39. **(a)** Static friction
Explanation: The movement in a stationary body is possible only when the applied force is greater than the frictional force. The frictional force that counter balances the applied force is called static friction. In static friction both the surfaces interlock each other strongly.
40. **(c)** Decrease friction
Explanation: Carpenters while fixing screws, grease the tips of screw to decrease the friction. Grease present on screw smooth the irregularities hence reduce the friction offered by wood articles.

41. **(d)** polished marble surface
Explanation: The car will travel farthest on the surface which will exert minimum friction. Therefore, a marbled surface will be much suitable for the 's movement.
42. **(c)** increase friction
Explanation: The handle of a cricket bat is usually rough to increase friction to have a better grip, therefore it doesn't slip out in the hand of the cricketer while batting.
43. **(b)** Rolling friction is less than sliding friction
Explanation: Ball bearing is used in cycles because rolling friction is less than sliding friction. Hence, most of the moving parts of machine have ball bearing to reduce friction.
44. **(c)** decrease
Explanation: Mobil oil is applied on iron shutters to decrease the friction. After the applying of mobil oil, the shutter become very smooth and we could easily de-shutter it by applying less effort.
45. **(a)** Unnecessary expenses of energy
Explanation: Friction involves unnecessary expenses of energy because due to friction heat is generated which leads to more consumption of energy or fuel.
46. **(d)** Surface irregularities on them
Explanation: The friction between any two objects is due to surface irregularities on them. These irregularities creates interlocking of the surfaces in contact to create friction.
47. **(c)** Reduce the efficiency of machine
Explanation: In moving machine, friction decreases the efficiency of machine. Friction in machine causes its heating due to which it smooth the surfaces causing damage to machine.
48. **(d)** Two bodies in contact are rough
Explanation: The friction is more when two bodies in contact are rough. Rough surface have more irregularities that interlock the surfaces and leads to more friction.
49. **(a)** Conduction
Explanation: The process of heat transfer in which particles transfer heat one after the other in contact is called conduction. The heat in metals is conducted by the process of conduction because molecules in solid are tightly packed. they just transfer their energy to adjacent molecule without leaving their place.
50. **(c)** Using matchstick
Explanation: Frictional force is necessary in using matchstick. When matchstick is strike with matchbox friction is caused due to which heat is generated. The phosphorous in matchstick catch fire.
51. **(b)** Reduce friction
Explanation: Sliding friction is less than static friction. Powder fill the rough surface of carrom board due to which friction between the carrom coins and the board reduces and coins slides easily.
52. **(a)** When sliding
Explanation: Sliding friction is always more than rolling friction because during sliding of a body more surface area is in contact not in case of rolling
so, a body rolling over the ground will offer less friction than body sliding over the same ground.
53. **(b)** Giving suitable shape
Explanation: Liquid and gases are called fluid. Friction in fluid can be minimized by giving suitable shape called streamline

shape. As streamlined shape is narrow at the ends and broader in the middle due to which it reduces the fluid friction and helps an object to move forward easily.

54.

(d) Treading of tyres

Explanation: Friction is necessary for movement of vehicle on road. To increase friction treading of tyres is done. Regular oiling of engine, cleaning of vehicle, and less inflating of the tyres reduce friction.

55.

(b) Friction exerted by water is very small

Explanation: It is difficult to apply brakes to stop the moving ship because friction exerted by water is very small to stop the moving ship having very high momentum.

56. **(a)** stone

Explanation: Stone will exert more friction and reaction which helps to sharpen the blade of the knife.

57.

(c) Wearing due to friction

Explanation: The sole of shoes becomes flat after using few months because of wearing due to friction. There is friction between road and sole of shoe constantly which leads to flatness of the shoe sole with time.

58. **(a)** Sandpaper has rough surface to increase friction

Explanation: Sand papers are used to clean the metal surface to remove rust because sandpapers has a rough surface to increase the friction to removes the rust.

59. **(a)** Light

Explanation: Friction of air can be observed if a falling object is light. When a tiny paper or feather falls from height, it takes long duration due to friction of air on it.

60.

(d) Contact force acting in opposite direction

Explanation: Friction is a contact force acting in opposite direction of moving body. Friction is created when one body moves over the surface of another body in contact to each other.

61. **(a)** Newton

Explanation: The S.I. unit of force is Newton. All kinds of force are measured in terms of Newton which is equal to force that push a body of mass 1kg to a distance of 1meter.

62.

(b) decrease

Explanation: The friction will reduce as oil works as a lubricant it will decrease the friction between the hinge and the door.

63.

(b) Sliding friction

Explanation: When an object moves on a surface, resistance is offered and moving object slows down. To keep object moving with same speed a constant force is required which is the measure of sliding friction.

Section B

64.

(b) from left to right ($\rightarrow$)

Explanation: The friction will act as an opposite force so it will work on the opposite direction. ($\rightarrow$)

Section C

65.

(d) Opposite

Explanation: The direction of friction is always opposite to the direction in which the object is moving. This opposite force is useful in case of walking, cycling, etc but harmful in case of wearing and tearing of shoes.

66.

(b) Less

Explanation: Sliding friction is less than static friction. When an object slides its surface didn't get enough time to be in contact with the surface of other due to which it becomes easier to slide the object.

67.

(b) Relative motion

Explanation: Friction opposes the relative motion between two surfaces in contact. Frictional force acts in opposite direction of motion to slow down the speed of moving body.

68.

(b) Smoothness/ nature

Explanation: Friction depends on the smoothness/ nature of the surface. Smoother surface offer less resistance as there will be less irregularities in surface which reduce the friction. As irregularities of the surface make the surface rough or smooth.

69.

(d) Friction

Explanation: A force of friction acts when an object moves through air or water. Fluids have drag force which inhibits the movement of objects in fluids. Frictional force always works in opposite direction of motion.

Section D

70. **(a) Static > sliding >rolling**

Explanation: Rolling friction is always less than sliding friction and sliding friction is less than static friction. So, correct order is static friction, sliding friction, and rolling friction in decreasing order. In static friction more area of contact is there due to which it is very higher than the others.

Section E

71.

(c) Friction does not act on a ball moving through air.

Explanation: Friction acts on a ball moving through air.

72. **(a) static, sliding, rolling**

Explanation: Friction comes into play when irregularities present in the surfaces of two objects in contact get interlocked with each other. Static friction comes into play when we try to move an object which is at rest. Sliding friction comes into play when an object slides over the surface of another object. In sliding friction, the time given for interlocking is very small. Hence, interlocking is not strong. Therefore, less force is required to overcome this interlocking. Because of this reason, sliding friction is less than static friction. Similarly, the area of contact in case of rolling friction is smallest as compared to static or sliding friction. This area of contact changes gradually because of rolling. Hence, rolling friction is lesser than both static and sliding friction.

Section F

73.

(b) wet marble floor, dry marble floor, newspaper and towel.

Explanation: Force of friction depends on the nature of surfaces in contact. The rougher the surface, the more is the friction between the surfaces in contact and vice-versa. Roughness present in the given surfaces can be arranged in an ascending order as wet marble floor, dry marble floor, newspaper and towel. Hence, the correct sequence of these surfaces when arranged according to the increase in the force of friction acting on the car is—wet marble floor, dry marble floor, newspaper and towel.